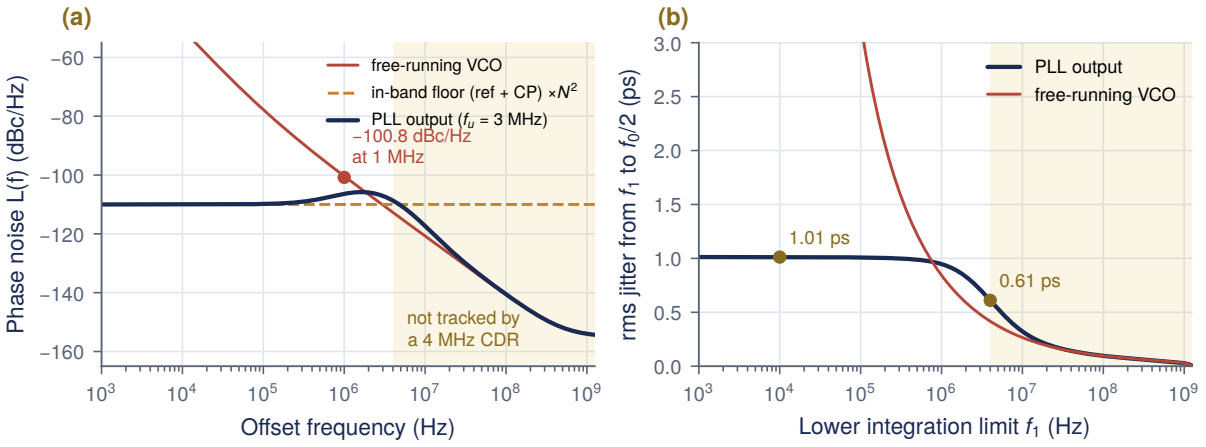
(a)
free-running VCO
in-band floor (ref + CP) ×N²
PLL output (f_u = 3 MHz)
−100.8 dBc/Hz
at 1 MHz
not tracked by
a 4 MHz CDR
Phase noise L(f) (dBc/Hz)
−60
−80
−100
−120
−140
−160
10³
10⁴
10⁵
10⁶
10⁷
10⁸
10⁹
Offset frequency (Hz)
(b)
PLL output
free-running VCO
1.01 ps
0.61 ps
rms jitter from f₁ to f₀/2 (ps)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Lower integration limit f₁ (Hz)